Vaja: Kaset

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1 Explain how your proposal fits the challenge(s)

a) How did you come up with your idea?

The idea emerged from building **Vaja AI** — a general-purpose AI cowork platform for Thai users. While developing the platform, a pattern became clear: every professional group faces the same core problem with AI tools — generic AI doesn't know their domain, speaks in formal language they don't use, and requires them to install yet another app.

This problem is most acute for Thai smallholder farmers. A farmer asking ChatGPT about tomato leaf blight gets a generic, English-biased answer. It doesn't know which fungicides are available at their local shop, doesn't understand the humidity patterns in Northern Thailand in April, and can't factor in current market prices at their regional market.

The gap became undeniable when researching agricultural extension services: **Thailand has approximately 1 agricultural extension officer for every 200+ farmers** in many provinces. These officers are overwhelmed — and farmers are making high-stakes decisions alone every day.

We realized that the Contextual Skills Engine already built into Vaja Kaset could solve exactly this: load specialized agricultural knowledge into the AI, deliver it through LINE OA (which farmers already use daily), and make it affordable through shared credit pools for farmer groups. No new app. No training required. Just chat in Thai.

b) Define your target users

Primary users: Thai smallholder farmers

- Rice, cassava, sugarcane, tomato, and longan growers in Central and Northern Thailand
- Typically own 2–15 rai of farmland
- Use LINE daily; minimal smartphone proficiency beyond messaging
- Make 5–10 high-stakes farm decisions per week with limited access to expert advice

Secondary users: Agricultural extension officers

- Use the web-based control room to monitor farmer conversations, spot recurring issues, and assist on complex cases
- Benefit by having AI handle routine advisory (pest ID, basic weather risk) so they can focus on field visits and complex problems

Potential B2B channel: Farmer cooperatives

- Can purchase shared credit pools for their members
- One group administrator manages the account; all cooperative members benefit

c) Key decisions your solution supports

Decision Category	Specific Decisions Supported
Crop management	Pest & disease diagnosis; treatment selection; prevention planning
Risk mitigation	Weather-based planting/harvesting timing; flood and drought risk alerts
Market decisions	Market price context; sell-now vs. hold guidance; price trend awareness
Farm records	Activity logging via conversation; cost and income summaries

d) Current challenges of the target group

1. Advisory gap at scale One extension officer cannot give timely, personalized advice to 200+ farmers simultaneously. Farmers default to guesswork, neighbor advice, or expensive agrochemical salesman recommendations.

2. Generic AI is not useful for Thai agriculture ChatGPT, Claude, and Gemini have no knowledge of:

- Thai crop varieties and regional planting calendars
- Pesticides and fertilizers available in Thai local markets
- Thai agricultural climate zones and seasonal risk patterns
- Regional market price dynamics (e.g., Chiang Mai longan vs. Bangkok wholesale)

3. App adoption barrier Rural Thai farmers are heavy LINE users but rarely install new apps. Every additional app install loses a large portion of potential users. Solutions that require a new app, account registration, or technical onboarding fail to reach the farmers who need them most.

4. Cost barrier for individuals AI subscriptions ($15–20/month) are unaffordable for smallholder farmers with seasonal income. A model where a farmer group shares costs is far more realistic than individual subscriptions.

5. Language and literacy Thai farmers communicate in informal, regional Thai. Some farmers have lower literacy and prefer speaking over typing. Solutions that require formal Thai commands, English prompts, or structured text inputs create friction that breaks adoption.

e) Present your solution

Solution: Vaja Kaset— AI Farm Advisor on LINE OA and Web chat

Vaja Kaset is an AI cowork platform with a **Contextual Skills Engine** — a system that loads specialized domain knowledge into an AI automatically, based on what the user is talking about. We have packaged this engine with an **Agricultural Skill Pack** (4 specialized skills) and delivered it through LINE OA — the messaging platform Thai farmers already use daily.

How the farmer experiences it: A farmer can interact with Vaja Kaset in whatever way is natural for them — type a question in informal Thai, **send a photo** of a sick plant for visual diagnosis, or **send a voice note** if typing is inconvenient. The system detects what topic they're raising and automatically loads the right domain knowledge into the AI. The farmer receives a structured, Thai-language response — formatted as a visual card in LINE, with follow-up suggestion chips they can tap — as if they had a farming expert in their pocket, available 24/7.

When a regional outbreak or weather risk emerges, extension officers can proactively **broadcast alerts** to all farmers on the channel at once.

crops, local products, and local conditions — as if they had a farming expert in their pocket.

The 4 Agricultural Skills:

Skill	What it does	Triggered by
pest-disease-consult	Structured diagnosis: identify disease/pest, severity, immediate action, prevention	Keywords: โรค, แมลง, ใบเหลือง, เน่า, จุด, etc.

weather-farm-risk	Real-time weather ⟶ farm risk interpretation: planting/harvesting timing, flood/drought alerts	Keywords: ฝน, แล้ง, อากาศ, ปลูก, เก็บเกี่ยว, etc.
crop-market-advisor	Market price context + sell/hold guidance for major Thai crops	Keywords: ราคา, ขาย, ตลาด, กิโล, บาท, etc.
farm-record-keeper	Activity logging via natural conversation; cost/income summaries on request	Keywords: บันทึก, ใส่ปุ๋ย, รด, ปลูก, เก็บ, ขาย, etc.

What differentiates Vaja Kaset from other solutions:

Differentiator	Other AI tools	Vaja Kaset
No new app required	❌ Need to install	✅ Works in LINE OA farmers already use
Photo-based disease diagnosis	❌ Text only	✅ Send a photo ⟶ visual AI diagnosis
Voice note input	❌ Text only	✅ Speak in Thai ⟶ transcribed + AI responds
Voice reply from AI	❌ Text only	✅ AI replies with both text and Thai voice audio

Thai agricultural domain knowledge	❌ Generic only	✅ Skill-specific, locally relevant knowledge
Natural informal Thai	❌ Formal/English-biased	✅ Informal Thai, regional dialect-tolerant
Group cost sharing	❌ Individual subscription	✅ Cooperative credit pools
Extension officer oversight	❌ None	✅ Web control room + broadcast alerts
Proactive farmer alerts	❌ Reactive only	✅ Officer broadcasts outbreak/weather alerts
Expandable expertise	❌ Fixed feature set	✅ New skills = new domain expertise

2 AI & Technical Approach

a) How AI is used

Type of AI: Large Language Models (LLMs) with a proprietary domain knowledge injection layer

Primary model: Google Gemini 3.1 Flash (optimized for Thai language quality and cost efficiency). The platform also supports 20+ models including GPT-4o, Claude, and others

The Contextual Skills Engine — how it works:

Farmer interacts via LINE (text / photo / voice note)

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LINE webhook ⟶ Vaja Kaset platform receives message

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Input type routing:

├── Photo ⟶ Vision model analyzes image ⟶ extracted description

├── Voice ⟶ Gemini Speech-to-Text ⟶ Thai transcript

└── Text ⟶ direct

↓

Skills Engine activates (using combined text + extracted description):

├── Keyword match: "โรค" / disease symptoms in photo ⟶ pest-disease-consult

├── Keyword match: "ฝน" ⟶ weather-farm-risk

└── LLM scoring: model rates skill relevance ⟶ top skills auto-activate

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Agricultural domain knowledge injected into AI context (3 tiers):

Tier 1: Skill catalog — LLM sees all available skills

Tier 2: Active skill instructions — structured diagnostic workflow, Thai crop knowledge

Tier 3: Resource files — Thai crop disease database, local pesticide guide

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LLM generates Thai-language response with domain expertise

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Response formatted and delivered:

├── Flex bubble card (structured visual layout for diagnoses)

├── Quick reply chips (3 follow-up suggestions the farmer can tap)

└── Voice reply (Thai TTS audio sent via push message, fire-and-forget)

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Conversation + metrics logged ⟶ Extension officer reviews via web control room

What problem does AI solve specifically? The LLM acts as the reasoning and communication layer. Without the Skills Engine, an LLM gives generic, low-value answers to Thai agricultural questions. With the Skills Engine, it follows a structured diagnostic workflow (ask about symptoms, crop, weather, spread pattern), references Thai-specific crop diseases and products, and delivers practical recommendations in natural Thai.

AI Capability	Problem Solved
Vision model on plant photos	Farmer doesn't need to know disease names — just photograph the symptom
Thai speech-to-text	Eliminates typing barrier for low-literacy or field-condition use

Thai TTS voice reply	Farmer can listen to advice while working in the field, hands free
LLM with Skills Engine	Generic AI + agricultural domain knowledge = locally accurate recommendations
Contextual skill activation	No menus to navigate — AI automatically detects what help is needed

Value added compared to non-AI applications:

- A non-AI chatbot with fixed decision trees cannot handle the combinatorial complexity of crop × disease × weather × severity × product availability
- A fixed FAQ system cannot ask follow-up questions and refine a diagnosis
- The LLM handles the conversation naturally, while the Skills Engine ensures the knowledge it applies is locally relevant and accurate

b) Data inputs

Data Source	Type	Usage
Farmer's natural language messages	Real-time user input	Primary input — symptoms, observations, questions
Weather API	Real-time external data	Current conditions, 7-day forecast for farmer's region
Agricultural knowledge base (SKILL.md files)	Curated domain knowledge	Thai crop diseases, regional conditions, local pesticide market
Conversation history	Session memory	Multi-turn diagnosis; follow-up questions
LINE OA rich menu selection	User intent signal	Pre-loads relevant skill before conversation begins

c) Output

Farmers receive **natural Thai-language responses** delivered directly in their LINE chat:

For pest & disease consultations:

- Disease identification (Thai common name + scientific name)
- Severity assessment (low / medium / high)
- Immediate action steps (specific to locally available products)
- Prevention for next season
- Referral note to consult local extension officer for complex cases

For weather & risk:

- Current risk level (drought, flood, heat stress, frost)
- Recommended farm action (delay planting, prepare drainage, harvest now)
- 7-day outlook translated into farm decisions

For market decisions:

- Current price context for the crop/region
- Price trend direction
- Practical framing ("ข้อมูลเพื่อประกอบการตัดสินใจ — not financial advice")

For farm records:

- Confirmation of logged activity (date, crop, activity, quantity, cost/income)
- Weekly or monthly summaries on request

Proactive outputs (extension officer-initiated):

- **Broadcast alerts**: Officer pushes a regional pest outbreak warning or weather alert to all farmers on the channel at once — they receive it as a LINE push notification

For extension officers (web control room):

- Conversation logs per LINE user
- Analytics: message volume, top topics, active channels
- Ability to intervene or assist in ongoing conversations

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d) Development stage

Prototype

The core platform (Vaja Kaset) is fully operational with:

- ✅ Multi-model AI chat (20+ models including Gemini, GPT-4o, Claude)
- ✅ Contextual Skills Engine (create, trigger, inject domain skills)
- ✅ LINE OA integration (webhook, rich menu, broadcast messaging)
- ✅ Web control room with conversation history and analytics
- ✅ Credit system with group sharing capability
- ✅ Agricultural skill pack (pest-disease-consult, weather-farm-risk, crop-market-advisor, farm-record-keeper) — content being finalized with Thai crop disease database

The system is deployable. The remaining work is content refinement (agricultural knowledge quality), user flow/ui adjustment, and a pilot with an actual farmer group.

e) Incorporation of target group (Prototype Track)

Formal target group incorporation is planned for the help of **mentoring phase**, where 1:1 consultation sessions will be used to gather farmer and extension officer feedback and refine the proposal before Round 2 submission.

3 Practical Implementation

a) User interaction

Farmers: Access the solution entirely through **LINE(primary) and Web chat** — the messaging app most Thai farmers already use daily.

1. Follow the "Vaja Kaset" LINE Official Account @930gsczc or Web chat https://vaja-ten.vercel.app/agent/91EtcKbMl
2. A rich menu at the bottom of chat shows shortcut buttons:
    - "ถามโรคพืช/แมลง" ⟶ pre-activates pest-disease skill
    - "เช็คอากาศ" ⟶ pre-activates weather-risk skill
    - "บันทึกฟาร์ม" ⟶ pre-activates farm-record skill
3. Farmer interacts by:
    - **Typing** in natural, informal Thai
    - **Sending a photo** of a sick plant, field condition, or payment slip
    - **Sending a voice note** — speaks naturally; AI transcribes and responds with text + voice audio
4. Response arrives as a formatted card with tappable follow-up suggestion chips

No app download. No account registration. No commands to learn.

Extension officers: Access a web-based control room https://vaja-ten.vercel.app (browser on any device) to monitor conversations, view analytics, and assist on complex cases.

b) Infrastructure requirements

For farmers:

- Any smartphone (Android or iOS) with LINE installed
- Basic mobile internet connection (LINE messages are lightweight — 2G/3G sufficient)
- No additional devices, sensors, or hardware required

For extension officers (control room):

- Any device with a web browser and internet connection

Platform infrastructure:

- Cloud-hosted (Next.js + serverless Postgres) — no on-site servers
- LINE webhook for real-time message handling
- Weather API for real-time forecast data
- Cloud storage (Cloudflare R2) for voice reply audio files
- All infrastructure scales automatically with user growth

c) Ease of use

For farmers — zero technical requirements:

- Just use LINE as normal — no new accounts, no commands, no training
- Works with typing, voice notes, or photos — whatever is natural
- System adapts to however the farmer phrases their question — no fixed format required
- Voice reply means farmers can receive advice without reading — useful in the field
- Rich menu shortcuts mean even first-time users know what the bot can do

Accessibility note: For farmers with lower literacy, the rich menu buttons provide a shortcut so they don't need to type a full question — one tap pre-loads the right context.

Farmer profile	How they use Vaja Kaset
Comfortable with typing	Types question in informal Thai
Prefers speaking	Sends voice note — AI transcribes + responds with voice

Lower literacy	Taps rich menu button + sends photo of the problem
Tech-unfamiliar	Follows QR code ⟶ one button tap ⟶ gets help

4 Economic Value

1. Who benefits financially?

Beneficiary	Financial benefit
Smallholder farmers	Reduced crop losses from earlier disease detection; better market timing; lower cost per advisory compared to agrochemical salesman recommendations
Farmer cooperatives	Shared cost across members reduces per-farmer expense to a fraction of individual AI subscriptions
Agricultural extension officers	Increased capacity: AI handles routine advisory, officers focus on field visits with higher impact per hour worked
Agricultural value chain	Improved input purchasing decisions, reduced post-harvest losses from better timing

2. How is value created?

Increased yields: Earlier pest/disease detection reduces crop loss. Timely weather-based decisions (harvest before rain, delay planting during drought) protect yield.

Reduced costs: Correct disease diagnosis reduces unnecessary or incorrect pesticide use — a significant cost for smallholders. Market information reduces the likelihood of selling at the worst possible time.

Risk reduction: Weather risk interpretation translates meteorological data (which farmers may not know how to read) into direct farm actions. Flood and drought early warnings allow farmers to take protective action.

Time savings: 24/7 availability via voice or photo removes the dependency on office hours. Extension officers save hours answering repeated common questions — redirecting their expertise to complex cases.

Knowledge transfer at scale: One well-designed agricultural skill pack, once created and validated, can serve thousands of farmers simultaneously — a dramatic multiplier on agricultural expertise.

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3. Business model

Credit-based model with cooperative sharing:

Vaja Kaset uses a credit system where AI usage consumes credits. Credits can be purchased in packs:

Pack	Credits	Price (THB)	Per-farmer cost (50-member group)
Starter	100	99 ฿	—
Standard	500	399 ฿	~8 ฿ / member
Pro	2,000	1,299 ฿	~26 ฿ / member
Cooperative	5,000	2,999 ฿	~60 ฿ / member

The cooperative model: A farmer group or cooperative buys one credit pool. The group administrator distributes usage to all members. This makes the per-farmer cost as low as a few baht per day — affordable even for subsistence farmers.

B2B2C channel: Agricultural cooperatives, agri-input companies, or NGOs (e.g., GETHAC partners) can sponsor credit pools for farmer groups as part of their outreach programs.

Long-term: a **Skills Marketplace** where agricultural universities, GETHAC, and extension offices can publish verified agricultural skill packs — creating a community-maintained knowledge ecosystem.

5 Introduce yourself or the team

Team members

Field	Details
Name	Thirathat Thongkaew
Age	44
Educational background	Humanities and IT training
Occupation / Role	IT, Developer, and product designer

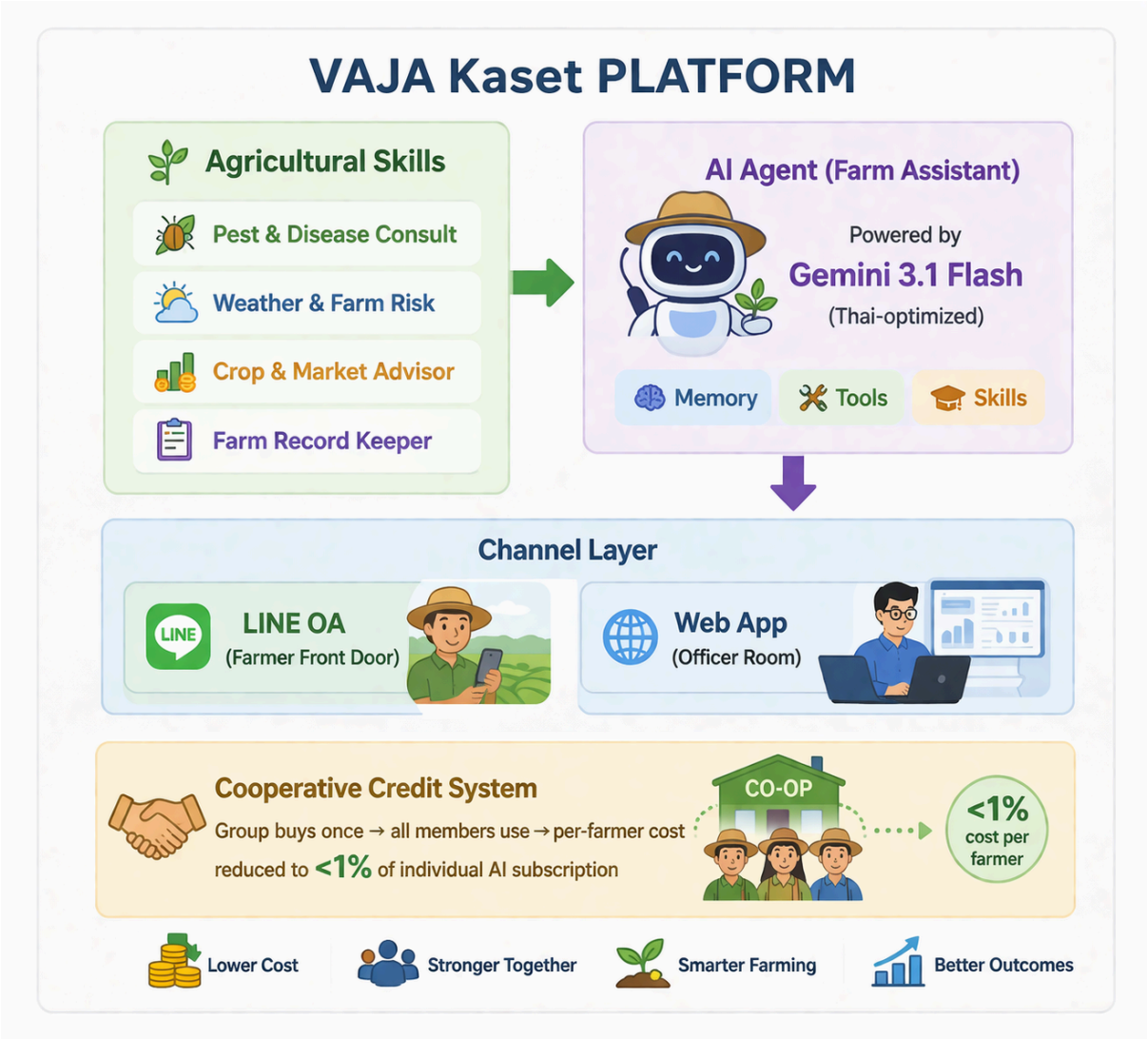
Vision-Mission Statement

Vision: Every Thai farmer has access to an agricultural AI advisor — in their own language, on the app they already use, at a cost they can afford.

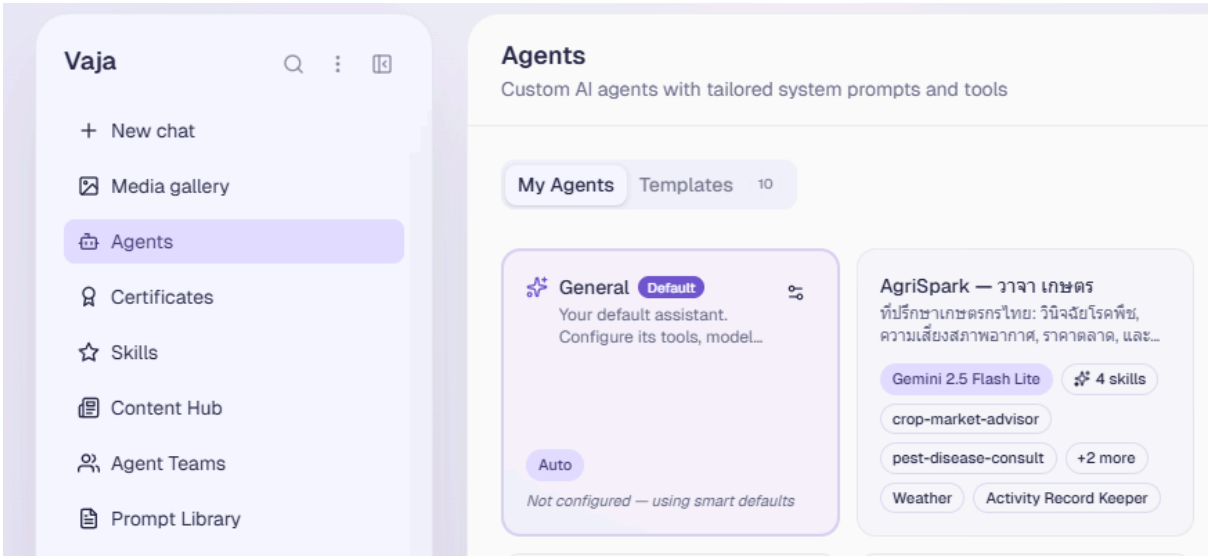
Mission: To build a skill-first AI platform that transforms domain expertise into accessible, affordable AI coworkers — starting with Thai agriculture, expanding to any profession that needs AI to truly understand their work.

6 Additional Material

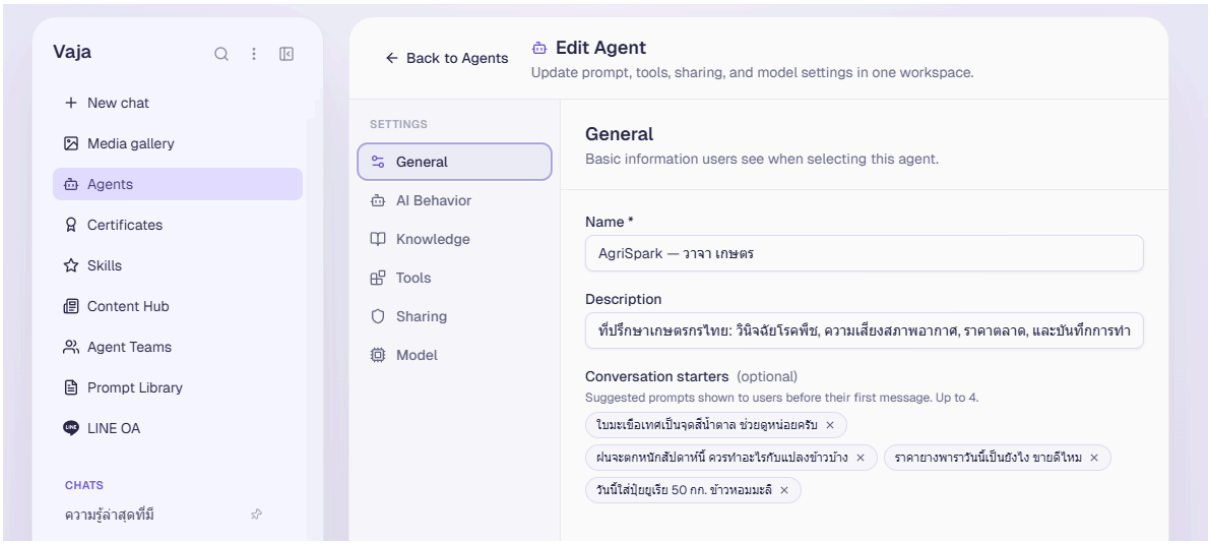
System Architecture Diagram



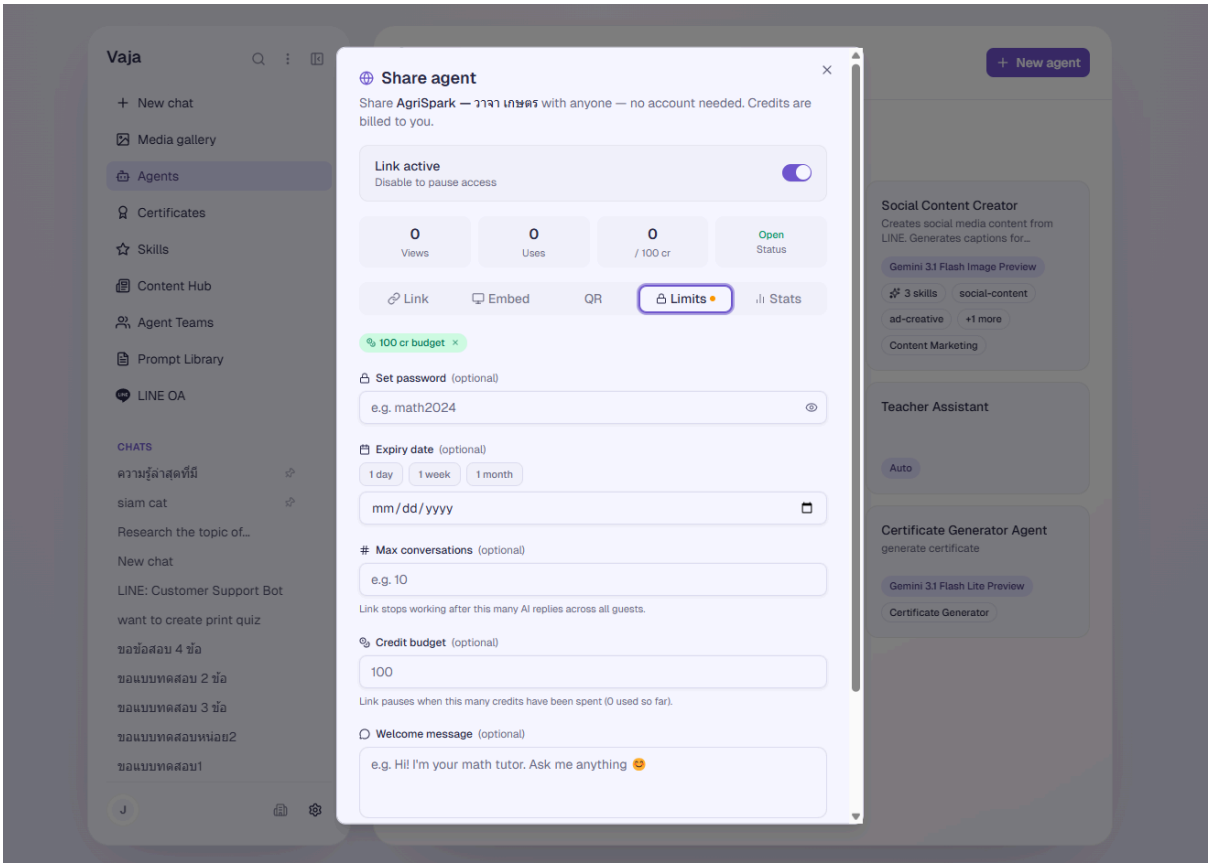
Agents Portal



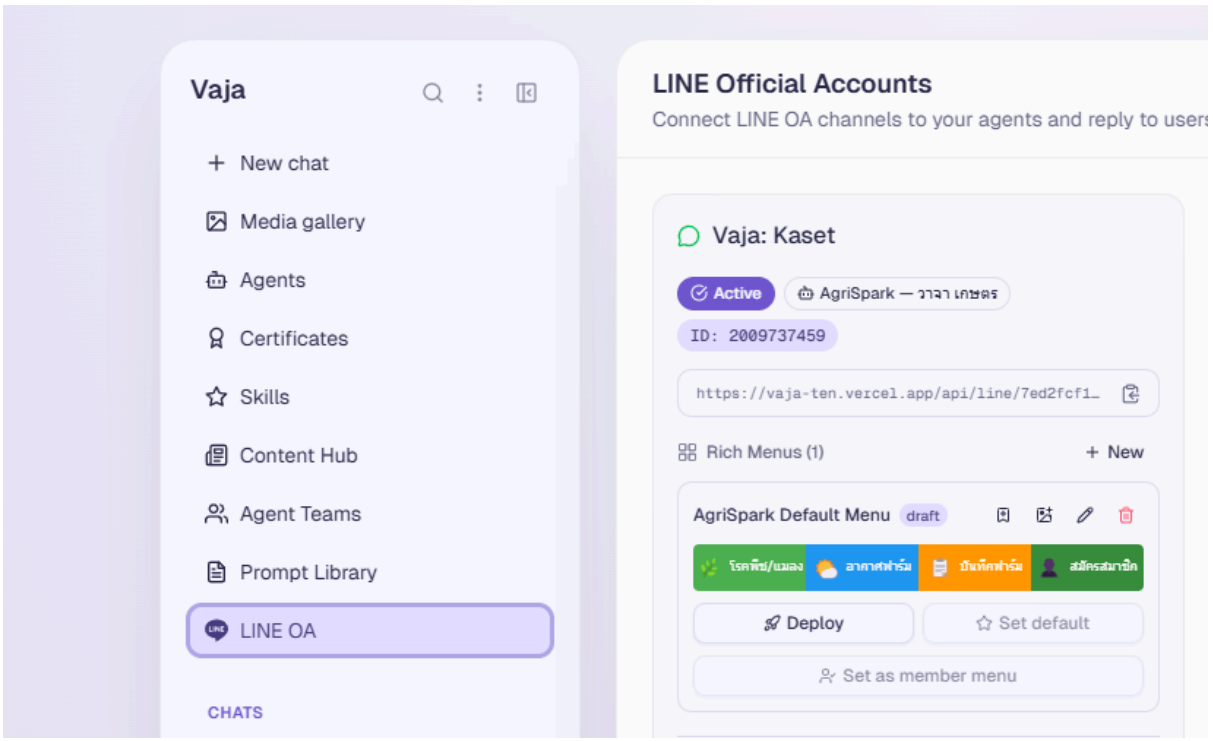
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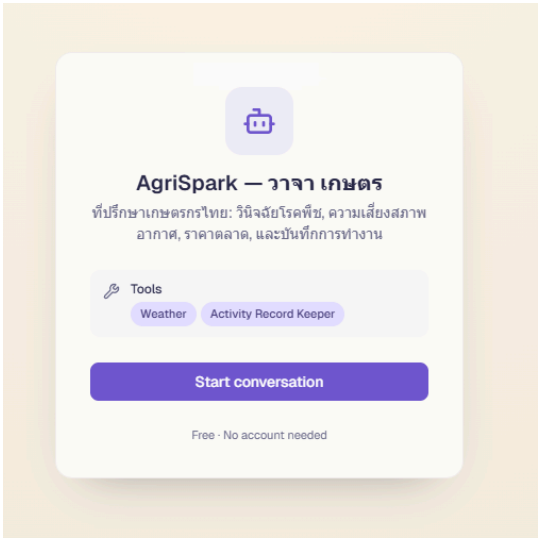
Agent Limit setting

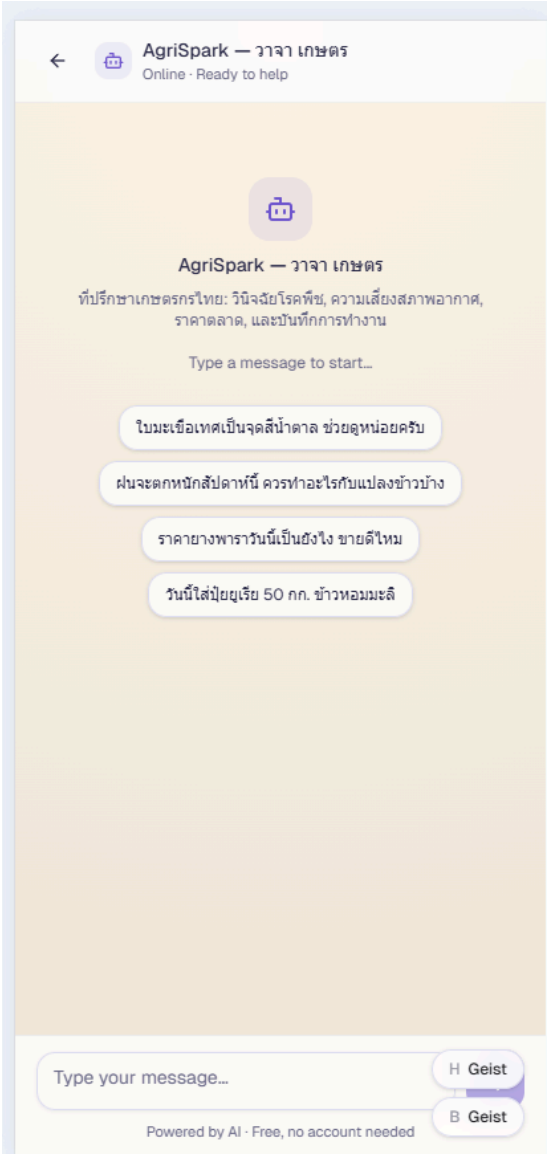


Line OA connection



Web Chat


AgriSpark — วาจา เกษตร
ที่ปรึกษาเกษตรกรไทย: วินิจฉัยโรคพืช, ความเสี่ยงสภาพอากาศ, ราคาตลาด, และบันทึกการทำงาน
Tools
Weather
Activity Record Keeper
Start conversation
Free · No account needed


AgriSpark — วาจา เกษตร
Online · Ready to help
AgriSpark — วาจา เกษตร
ที่ปรึกษาเกษตรกรไทย: วินิจฉัยโรคพืช, ความเสี่ยงสภาพอากาศ, ราคาตลาด, และบันทึกการทำงาน
Type a message to start...
ใบมะเขือเทศเป็นจุดสีน้ำตาล ช่วยดูหน่อยครับ
ฝนจะตกหนักสัปดาห์นี้ ควรทำอะไรกับแปลงข้าวบ้าง
ราคายางพาราวันนี้เป็นยังไง ขายดีไหม
วันนี้ใส่ปุ๋ยยูเรีย 50 กก. ข้าวหอมมะลิ
Type your message...
H Geist
B Geist
Powered by AI · Free, no account needed